

Lecture 3

CMOS Transistors & the Fabrication Process

Prof Peter YK Cheung

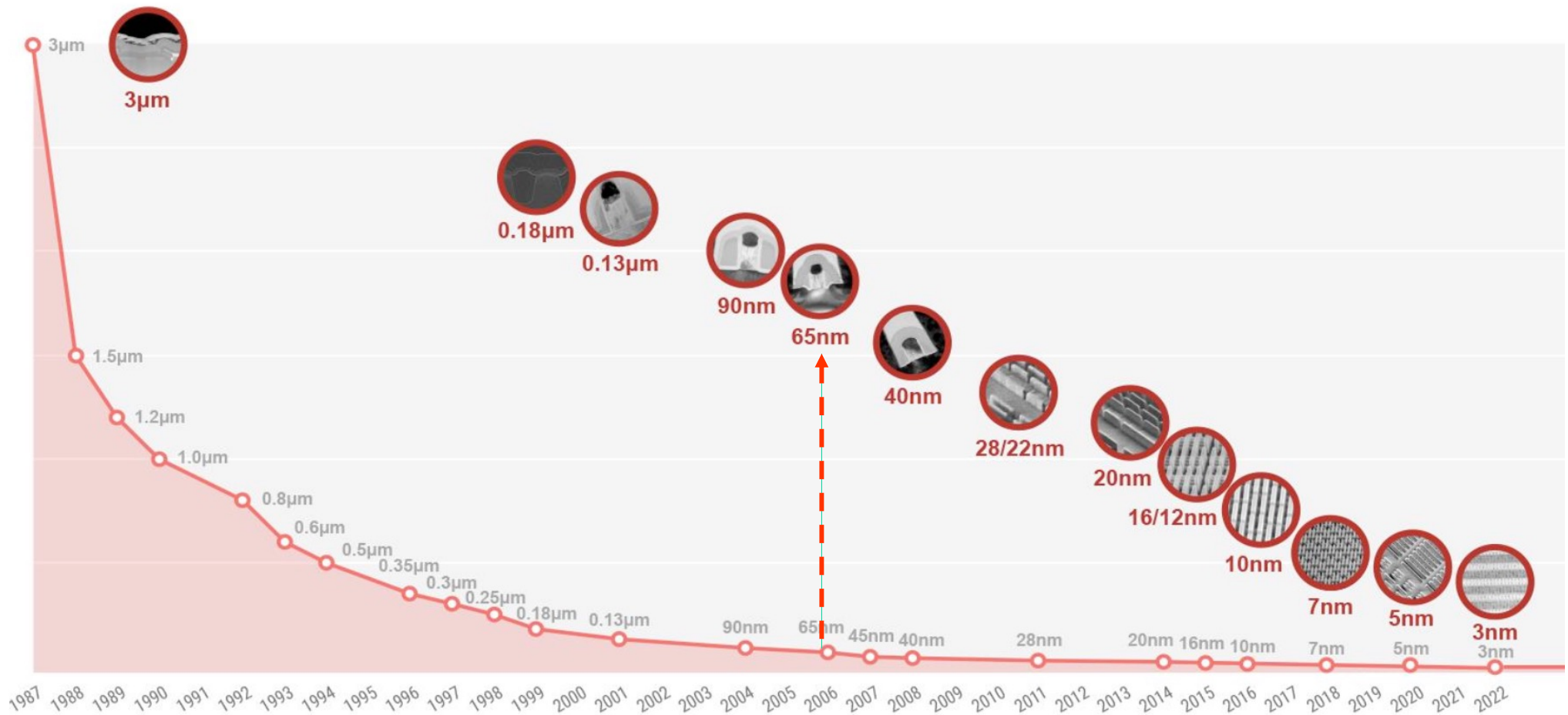
Department of Electrical & Electronic Engineering



URL: <https://elec70142-digital-vlsi-design.github.io/course-website/>

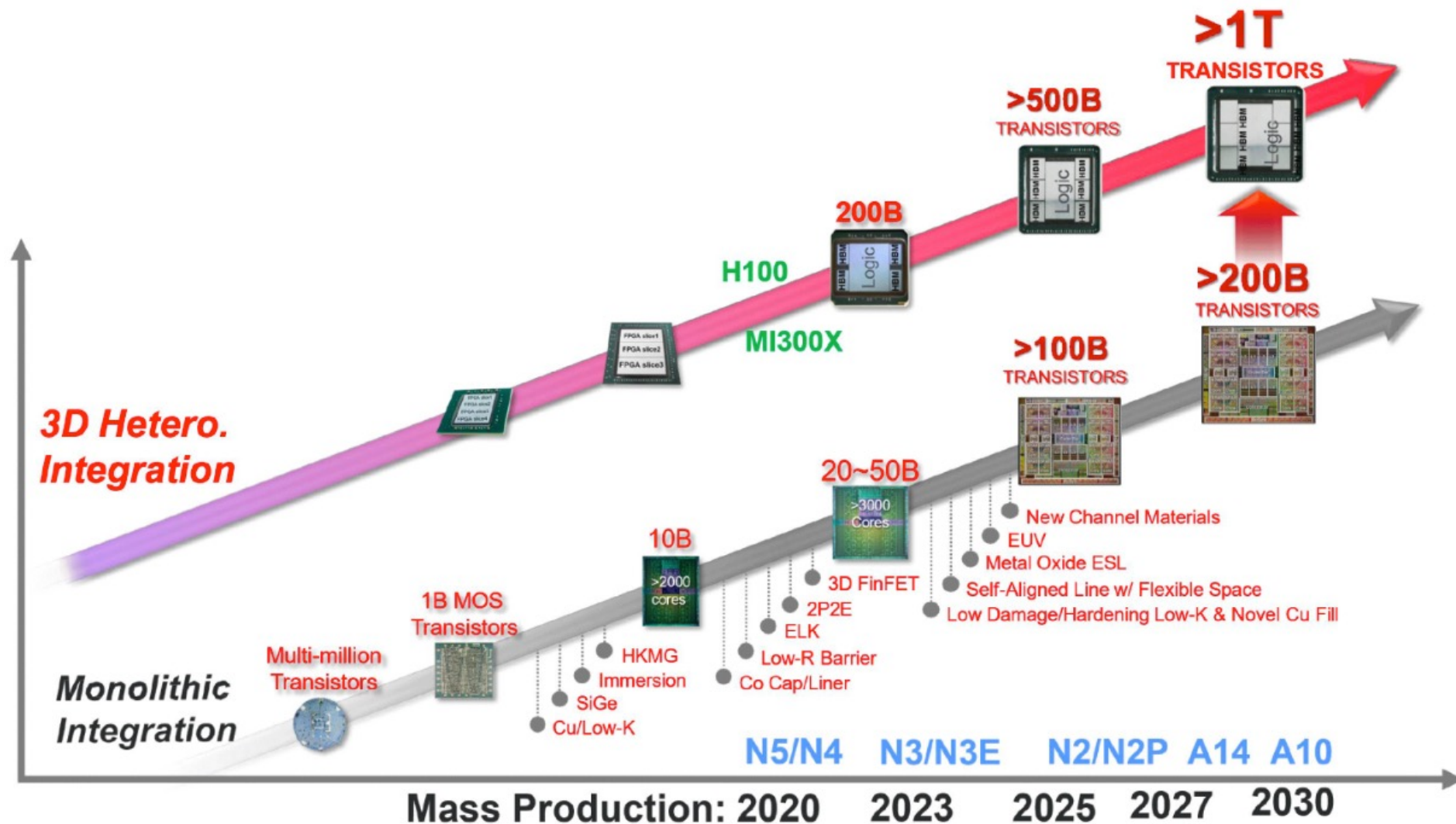
E-mail: p.cheung@imperial.ac.uk

TSMC CMOS Technology



Source: TSMC

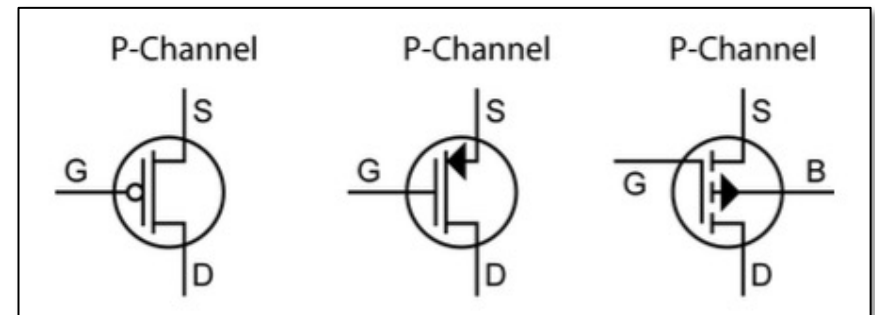
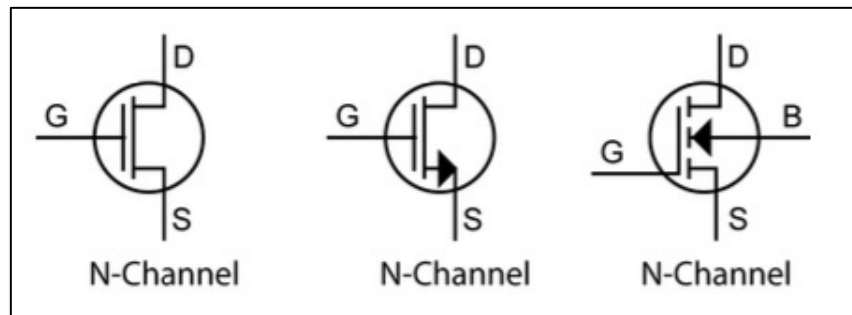
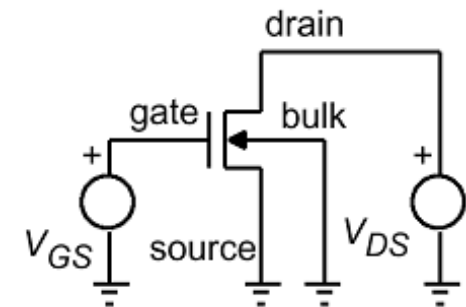
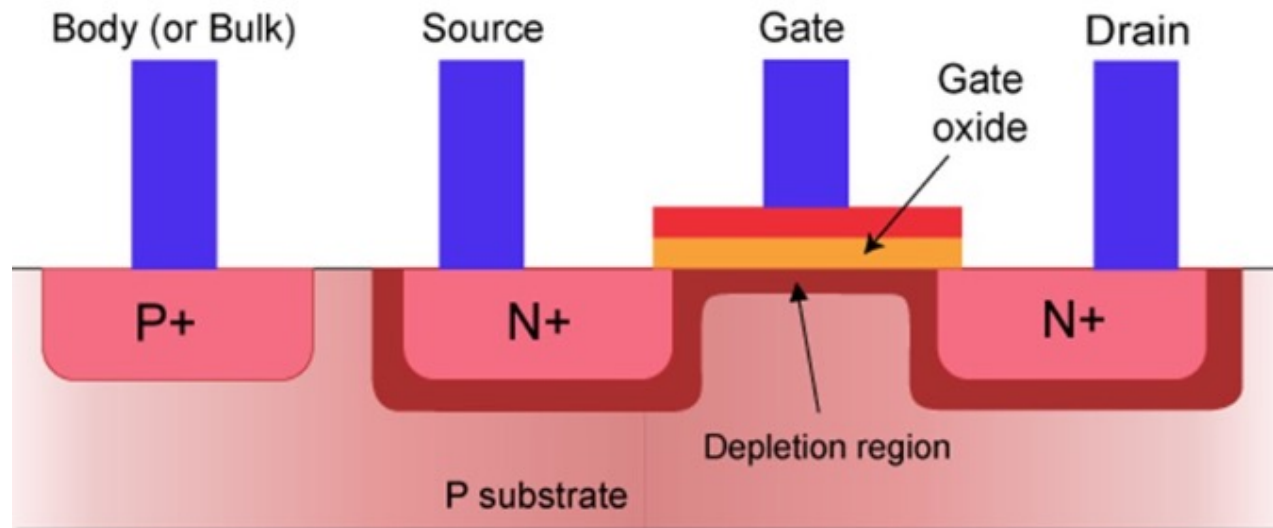
Semiconductor is going 3D



Source: Tom's Hardware

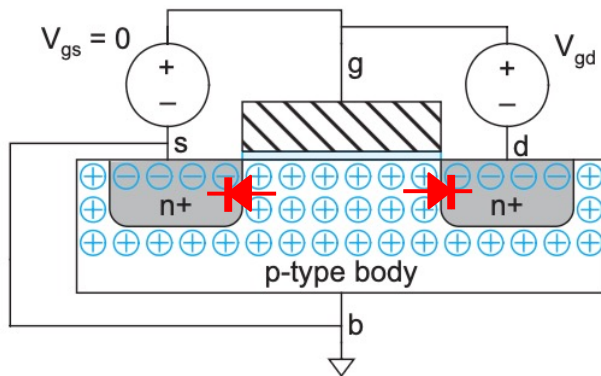
Basic nMOS Transistors

N-channel MOSFET transistor

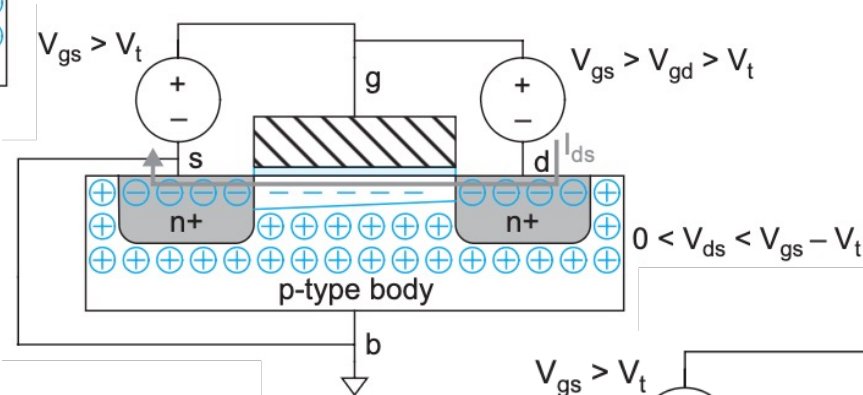


3 states of an nMOS transistor

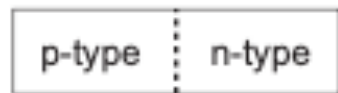
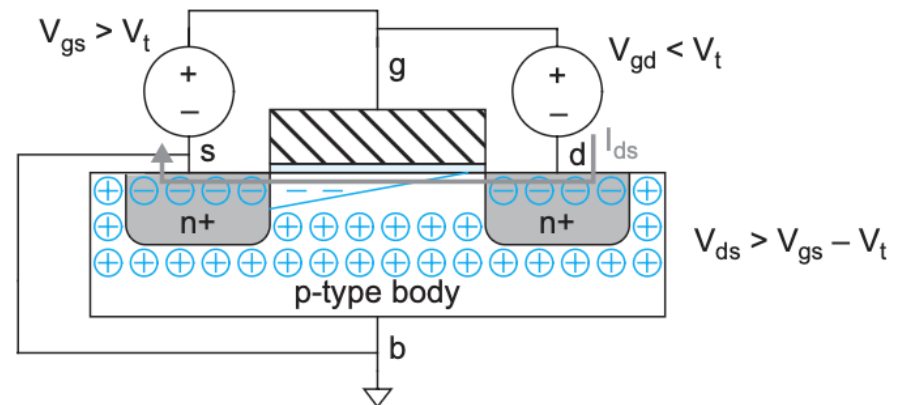
Cut off: no channel
 $V_{gs} = 0, I_{ds} = 0$



Linear: channel formed
 $V_{gs} > V_t, I_{ds}$ increase with V_{ds}



Saturation: Pinched off
 $V_{gs} > V_t, I_{ds}$ indep. V_{ds}

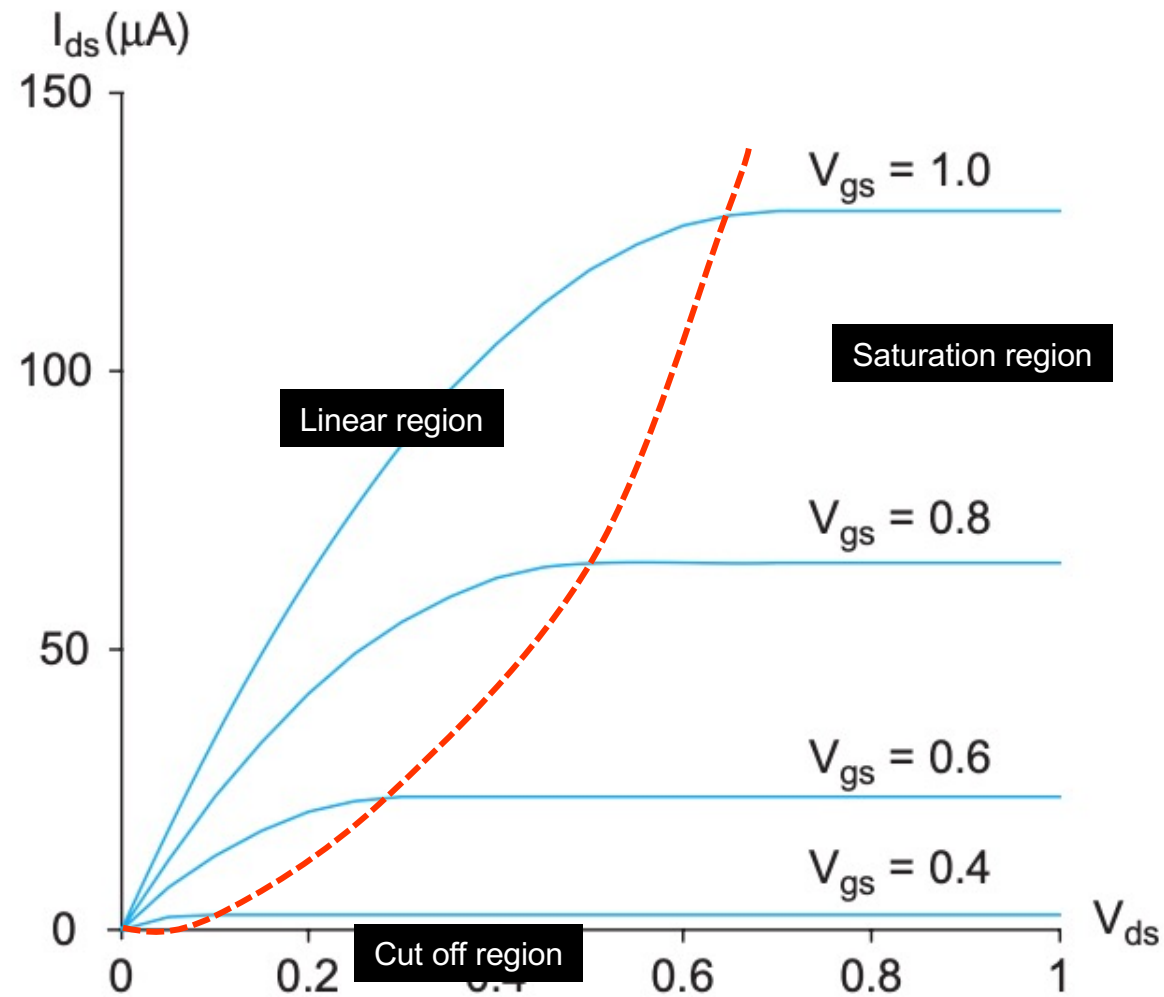


Anode Cathode

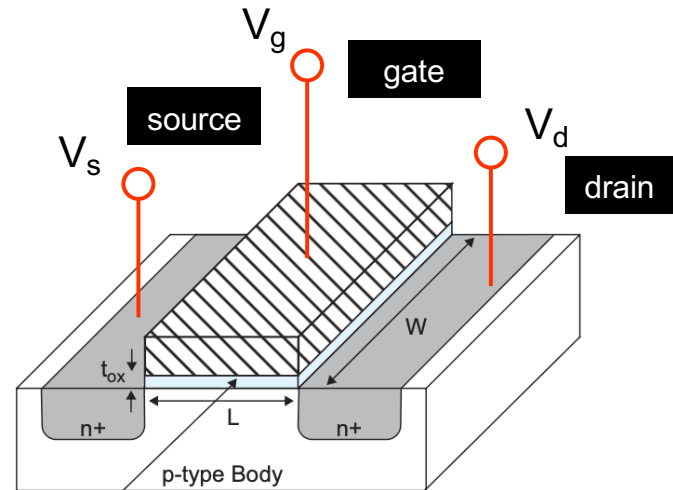


W&H p61-64

nMOS I-V characteristics



Key Transistor Equations for digital



SiO₂ Gate Oxide
(insulator, $\epsilon_{ox} = 3.9\epsilon_0$) ϵ_0 is the permittivity of free space, 8.85×10^{-14} F/cm

Key current equation

$$I_{ds} = \begin{cases} 0 & V_{gs} < V_t & \text{Cutoff} \\ \beta(V_{GT} - V_{ds}/2)V_{ds} & V_{ds} < V_{dsat} & \text{Linear} \\ \frac{\beta}{2}V_{GT}^2 & V_{ds} > V_{dsat} & \text{Saturation} \end{cases}$$

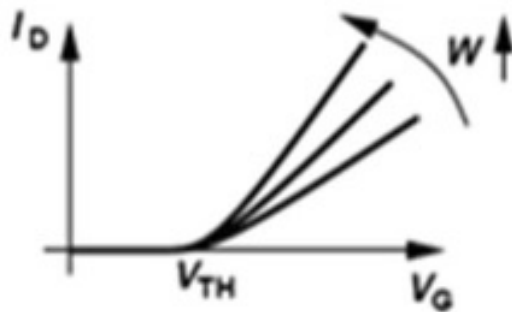
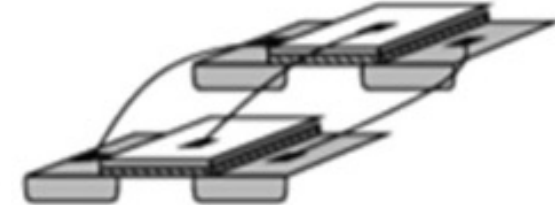
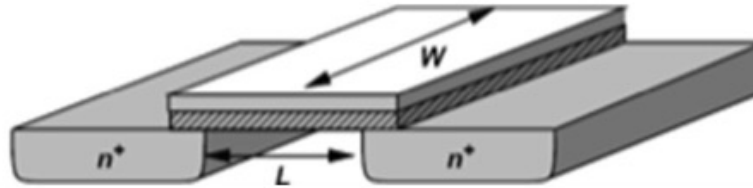
$$\beta = \mu C_{ox} \frac{W}{L}$$

$$V_{GT} = V_{gs} - V_t$$

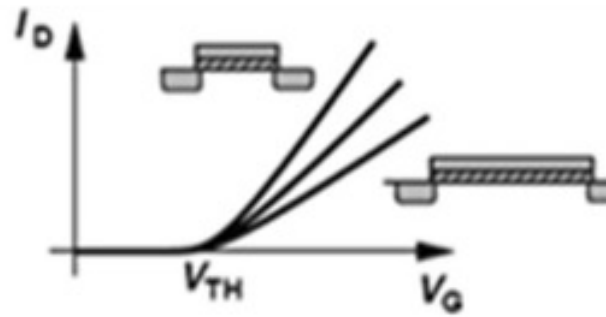
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Width, Length and t_{ox} dependency

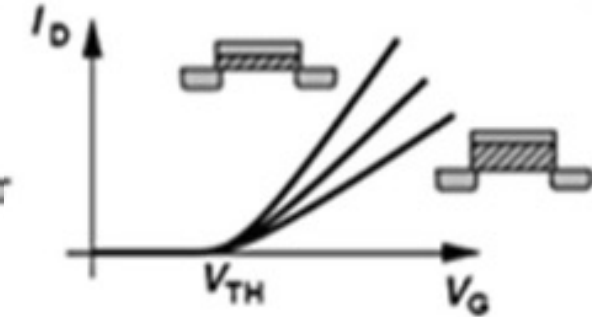
$$\beta = \mu C_{ox} \frac{W}{L}$$



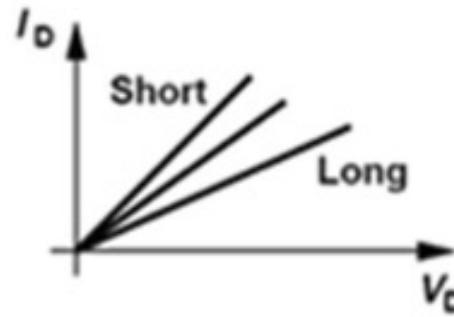
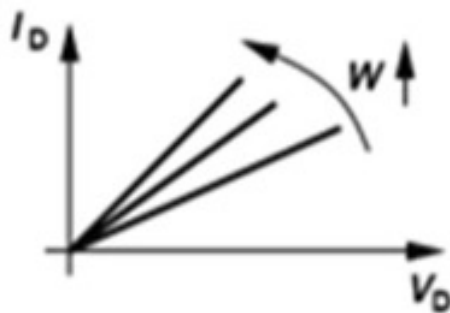
Width (W)



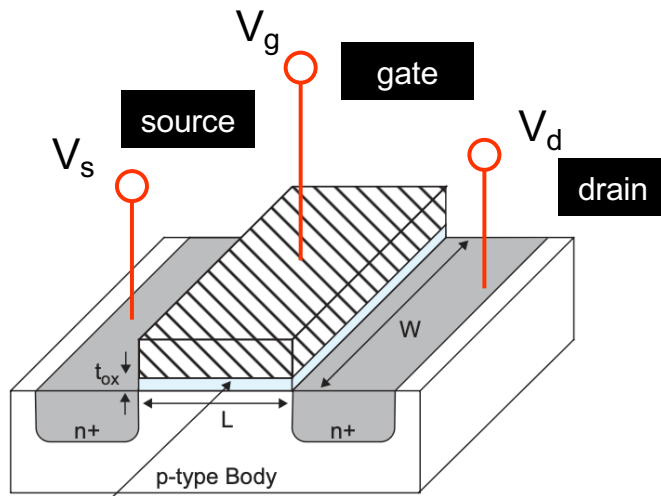
Length (L)



Gate oxide thickness (t_{ox})



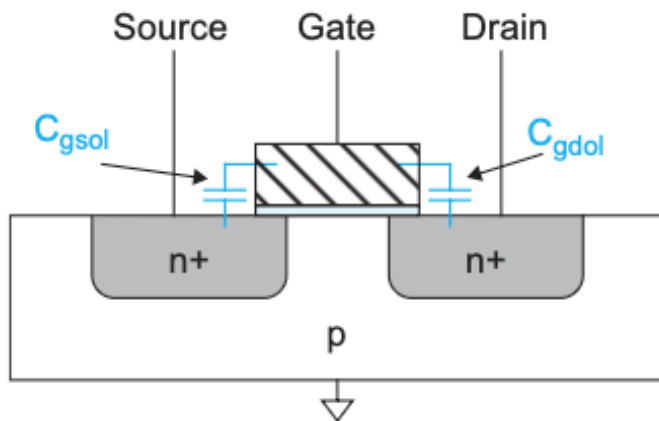
Capacitance Model



SiO₂ Gate Oxide
(insulator, $\epsilon_{ox} = 3.9\epsilon_0$) ϵ_0 is the permittivity of free space, 8.85×10^{-14} F/cm

- ◆ Gate capacitance as parallel plate:

$$C_0 = k_{ox} \epsilon_0 \frac{WL}{t_{ox}} = \epsilon_{ox} \frac{WL}{t_{ox}} = C_{ox} WL$$

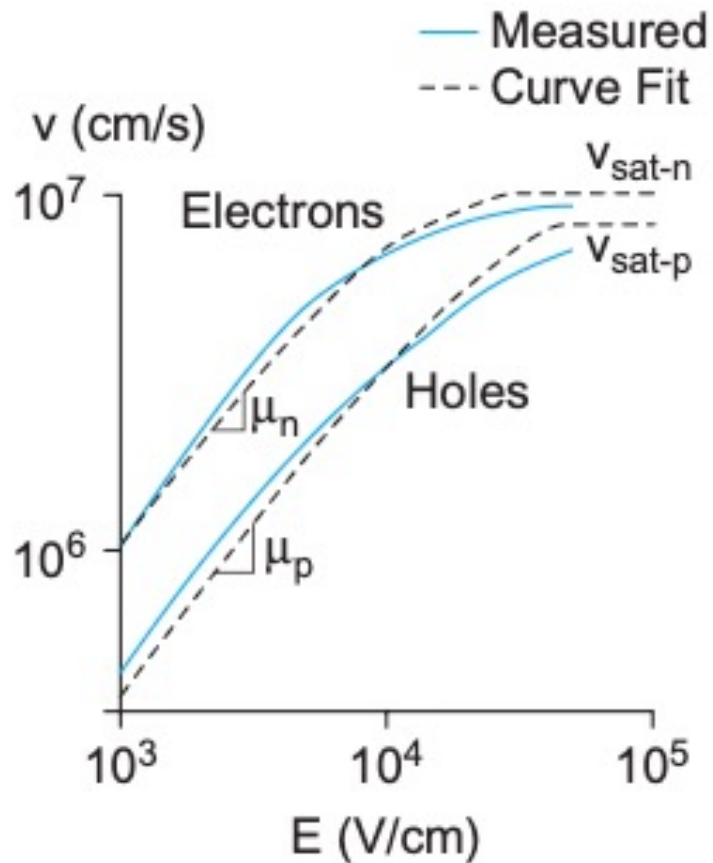


Parameter	Cutoff	Linear	Saturation
C_{gb}	$\leq C_0$	0	0
C_{gs}	0	$C_0/2$	$2/3 C_0$
C_{gd}	0	$C_0/2$	0
$C_g = C_{gs} + C_{gd} + C_{gb}$	C_0	C_0	$2/3 C_0$

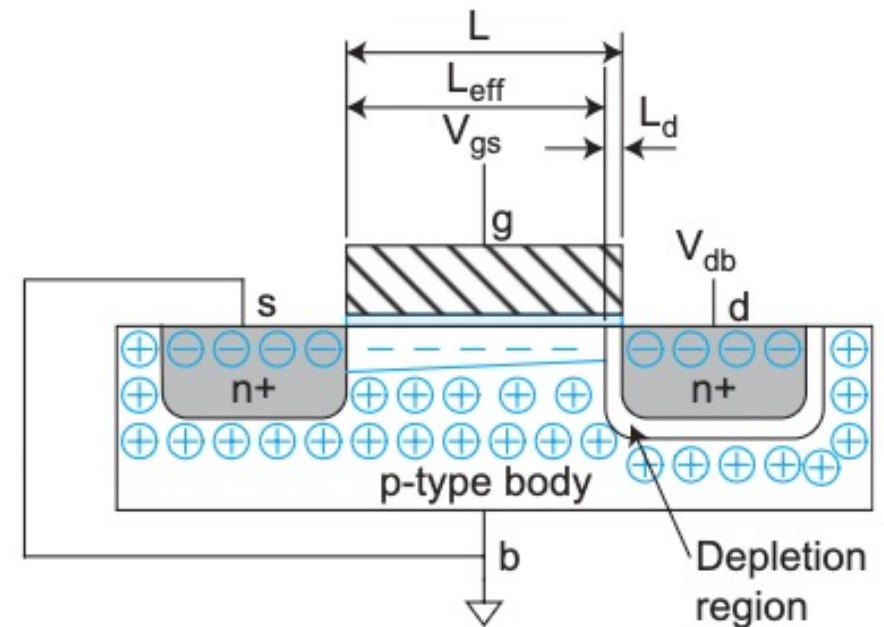
W&H p71-73

Non-ideal Effects (1) – on transistor gain

Mobility Degradation and Velocity Saturation



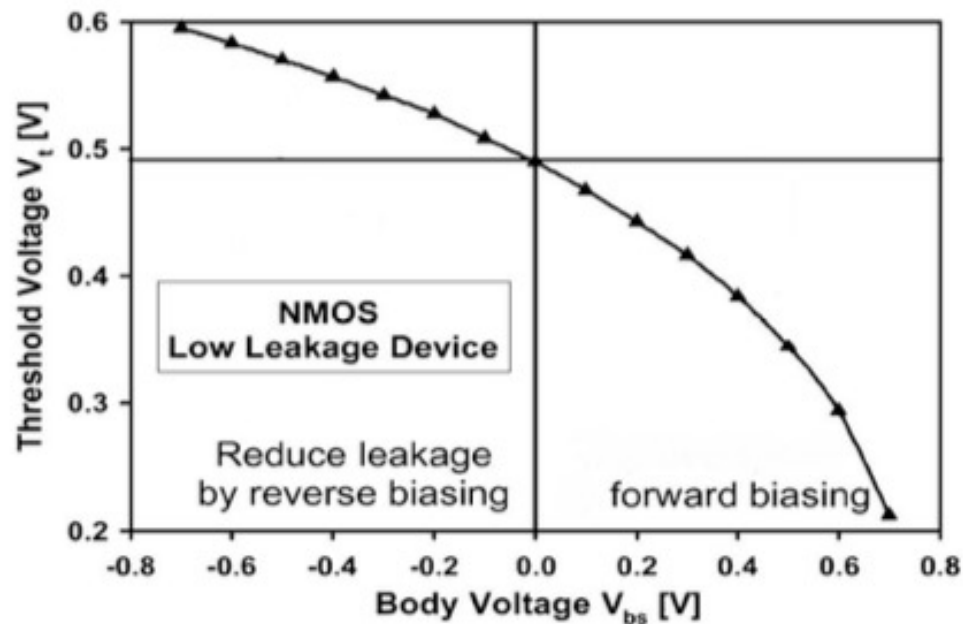
Channel Length Modulation



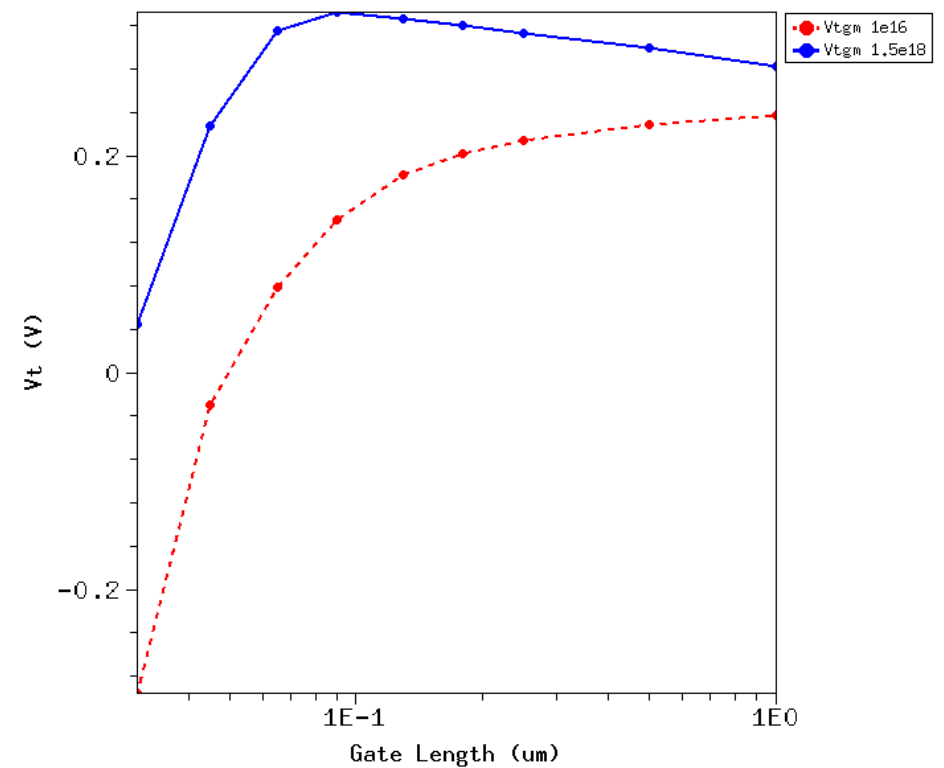
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Non-ideal Effects (2) – on threshold voltage

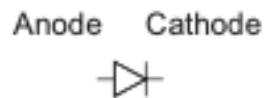
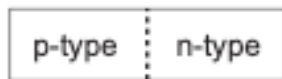
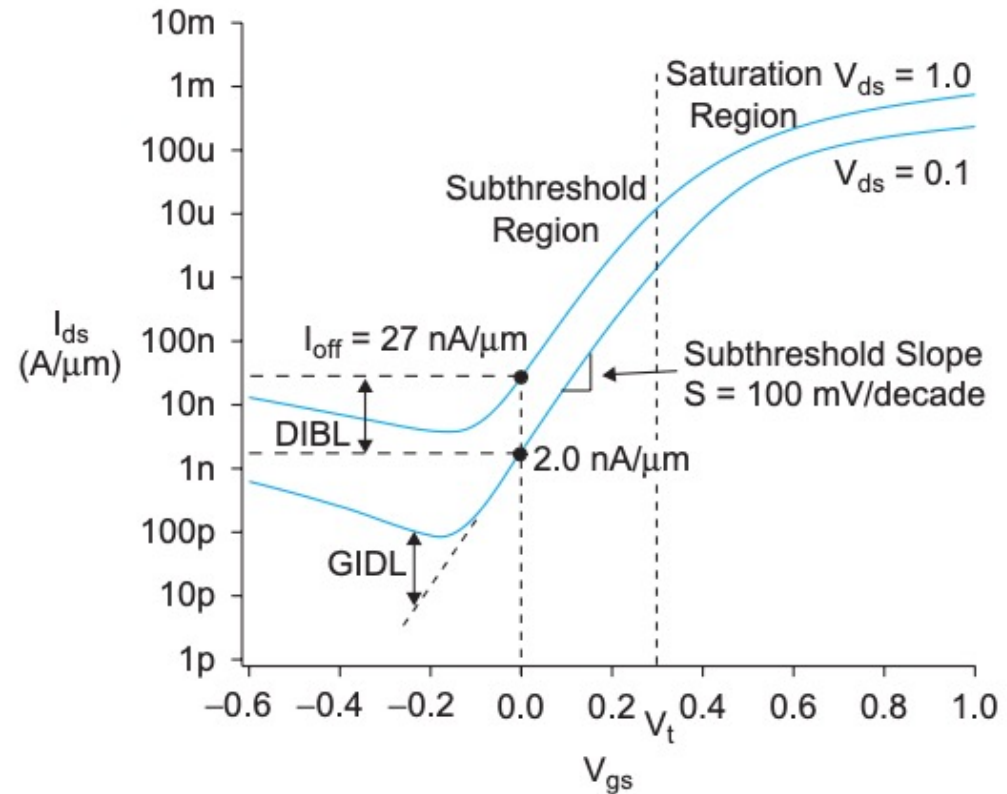
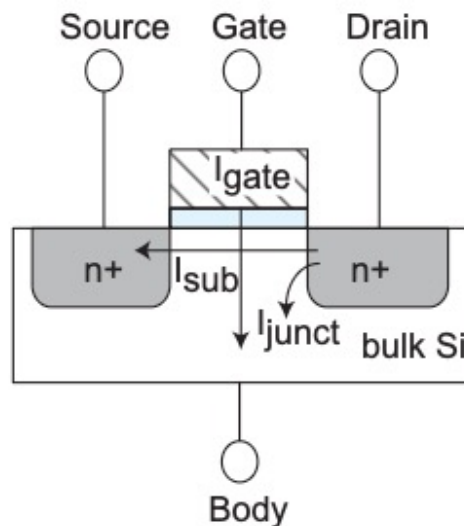
Body Effect



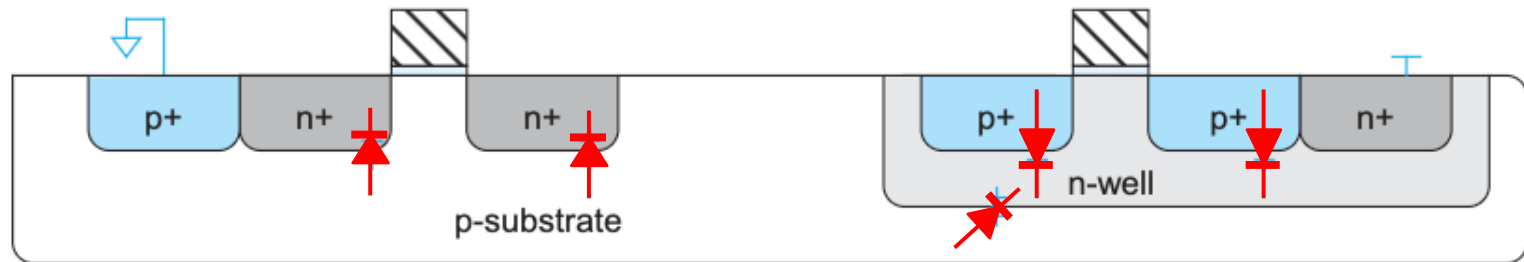
Short Channel effect



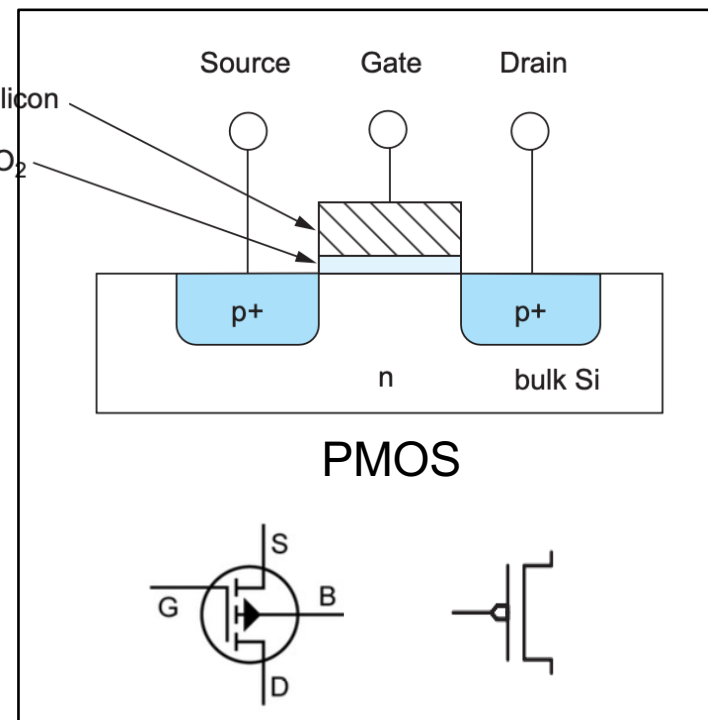
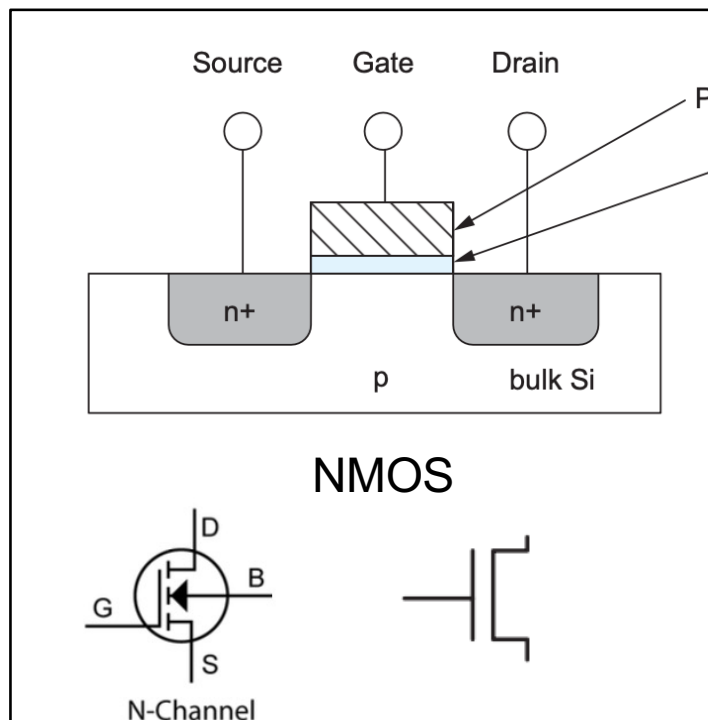
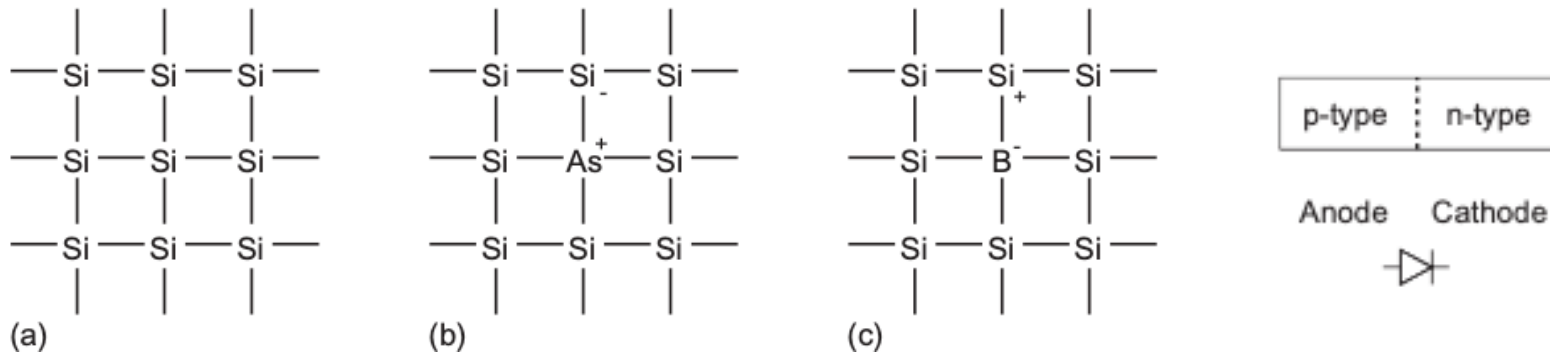
Non-ideal Effects (3) - leakages



W&H 80-84

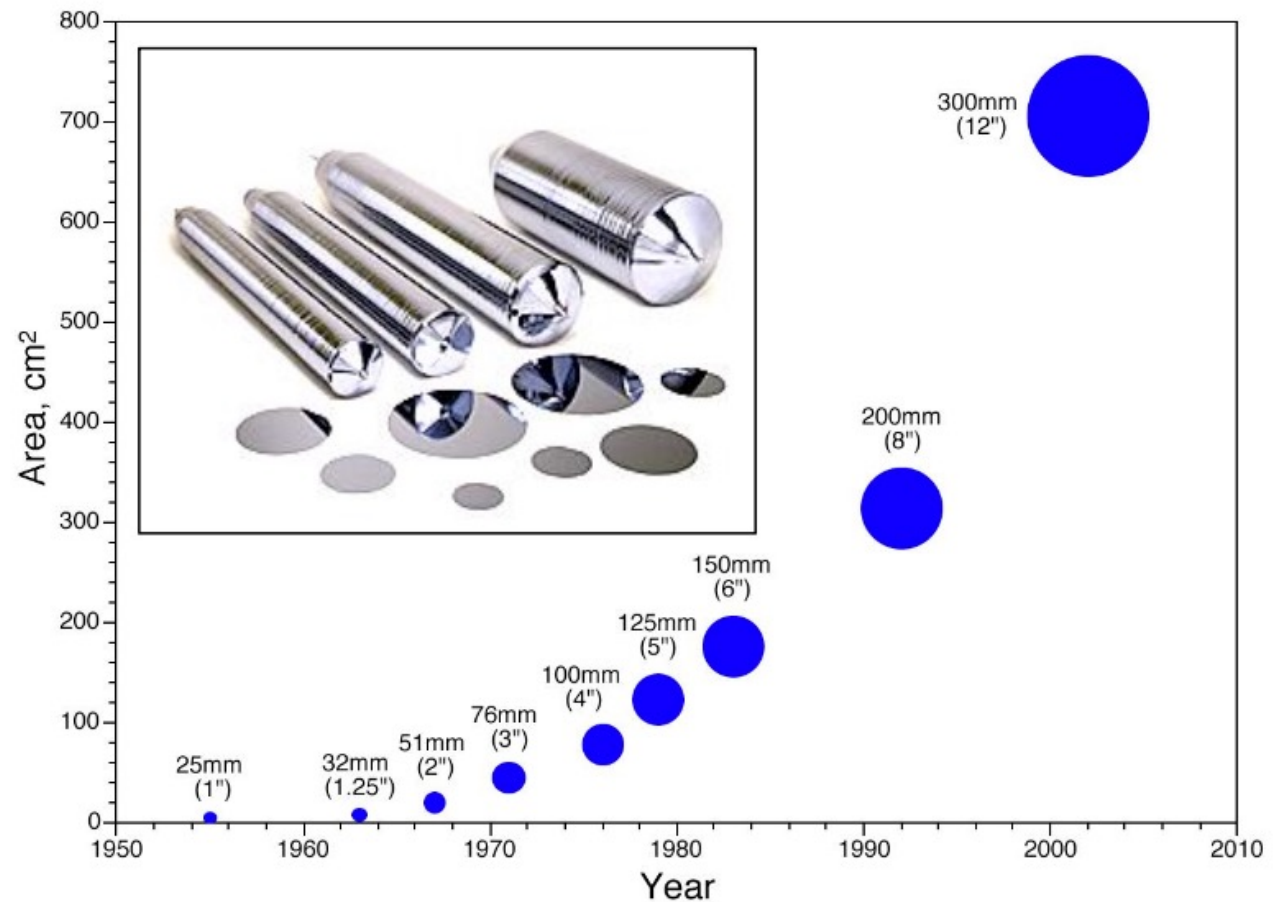
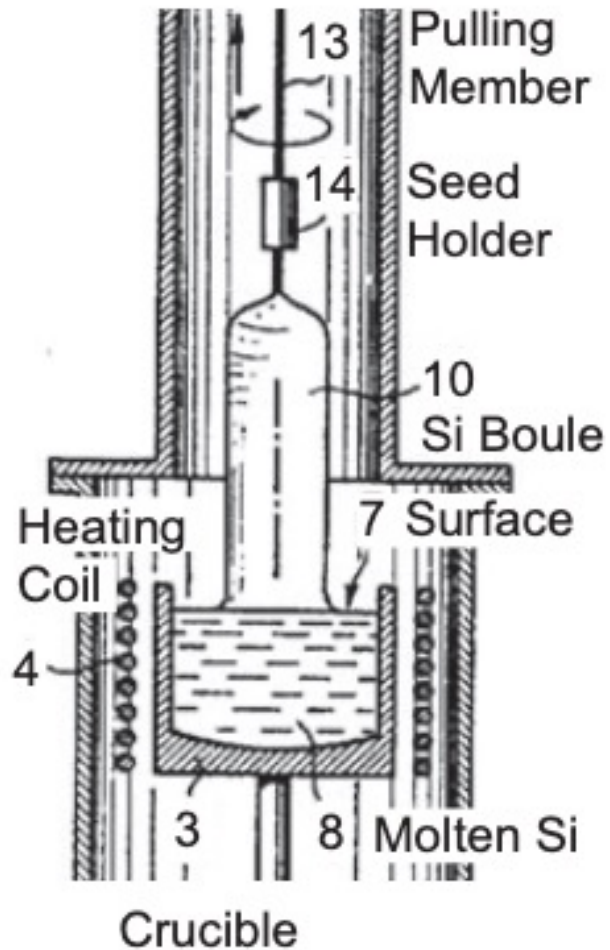


Substrate, n-type and p-type semiconductor

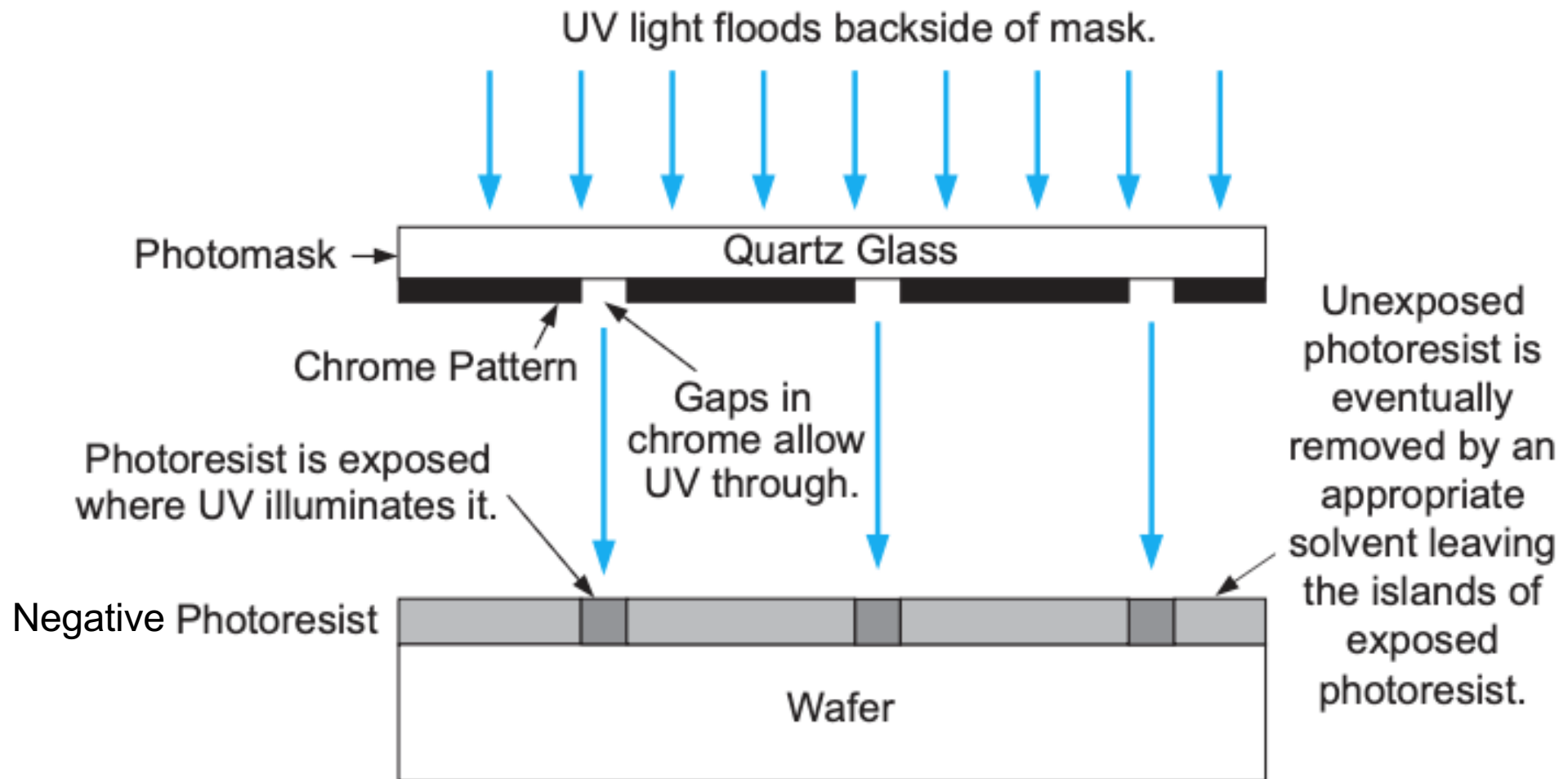


W&H p6-7

Silicon wafers

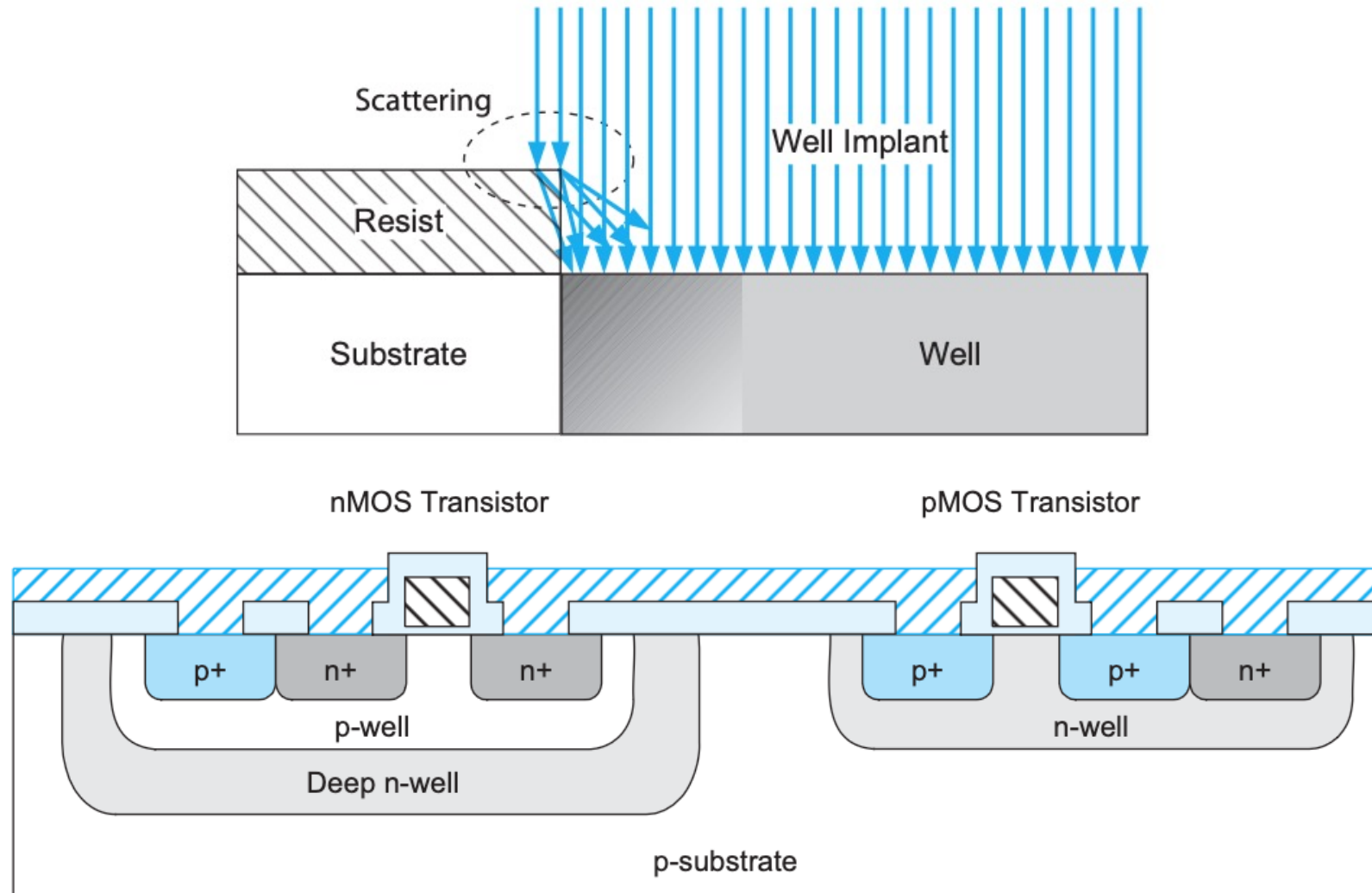


Photolithography



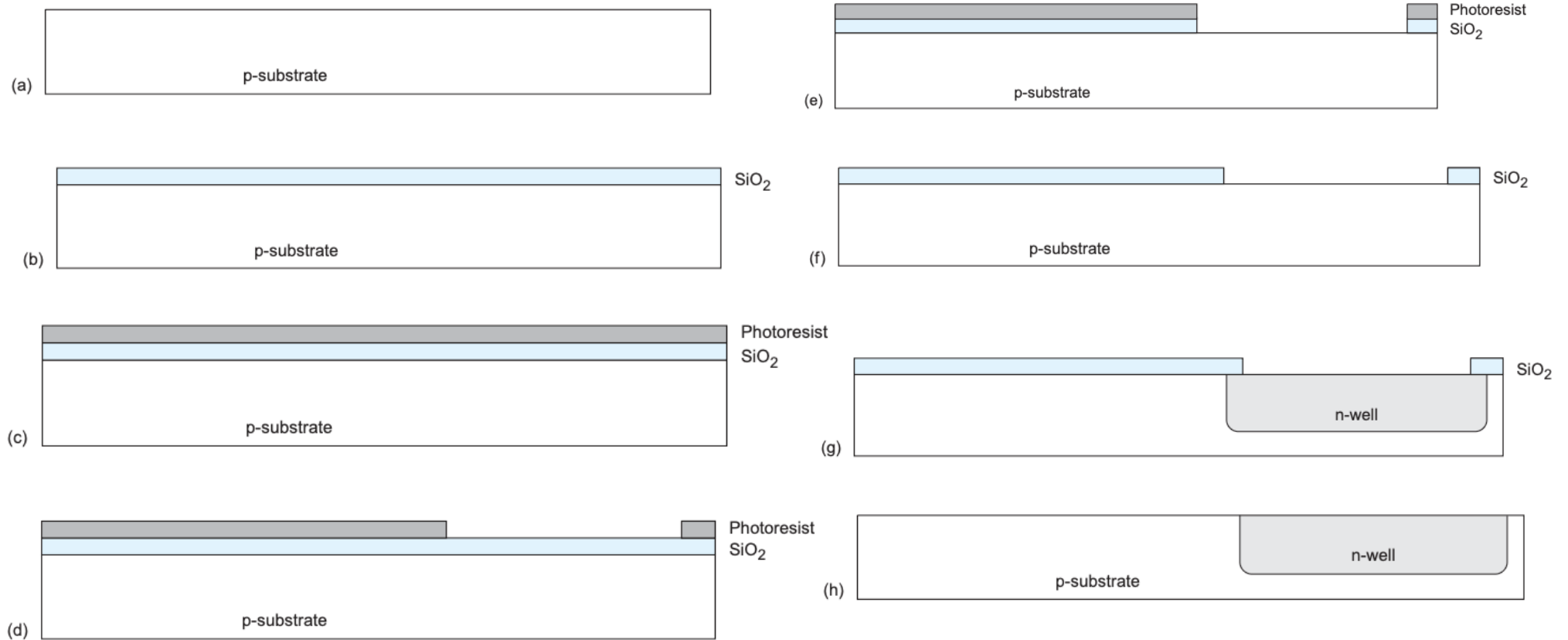
W&H p101-103

Well Implantation

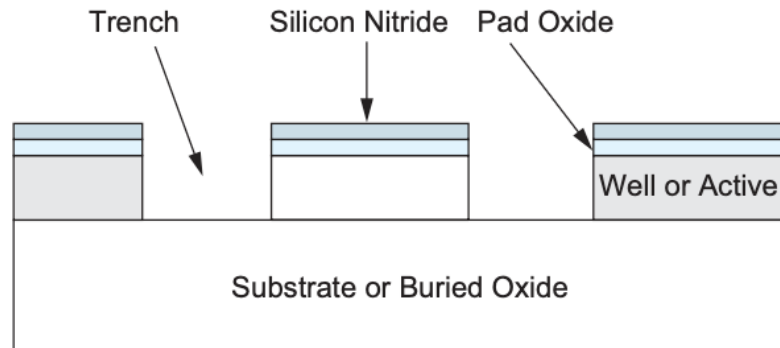


W&H p105

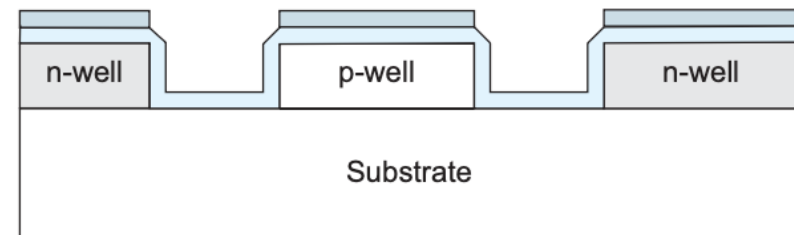
Photolithography in action – forming n-well



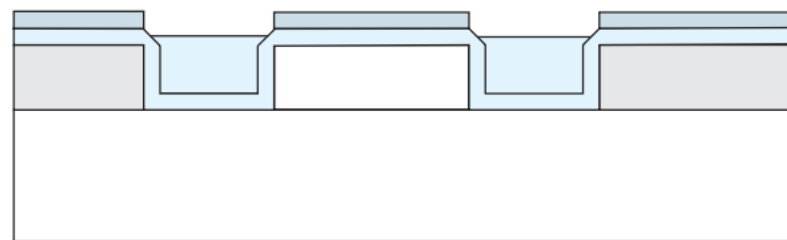
Shallow Trench Isolation (STI)



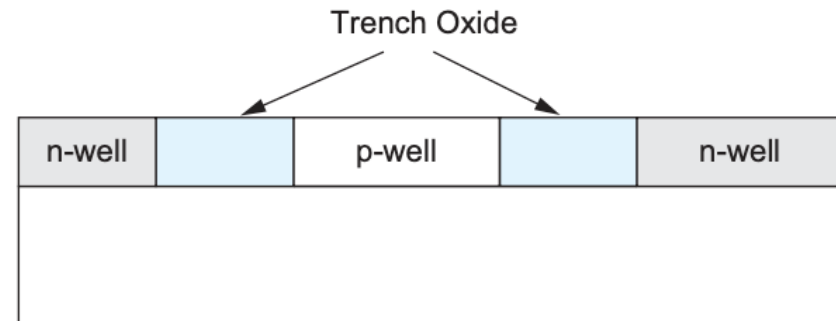
(a) Trench Etch



(b) Liner Oxidation

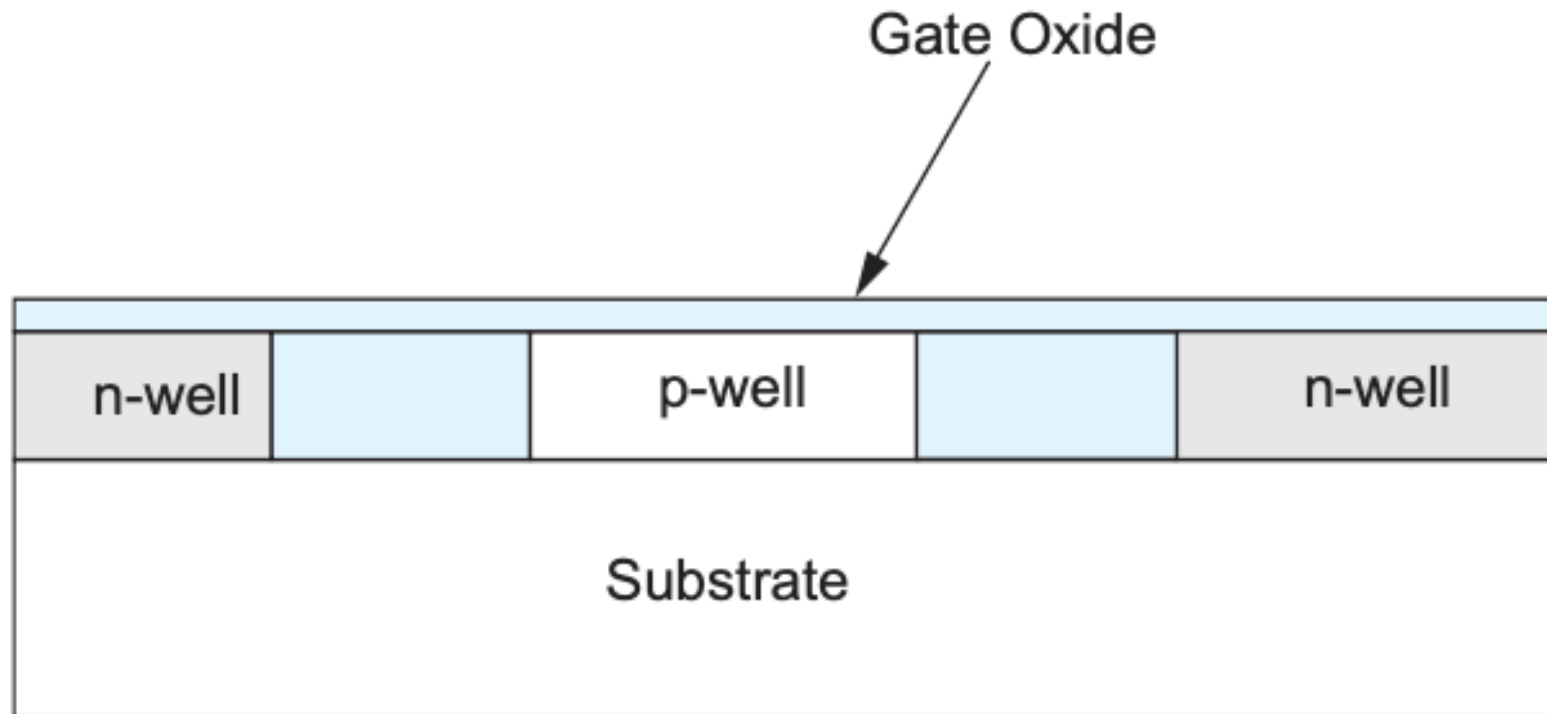


(c) Fill Trench with Dielectric

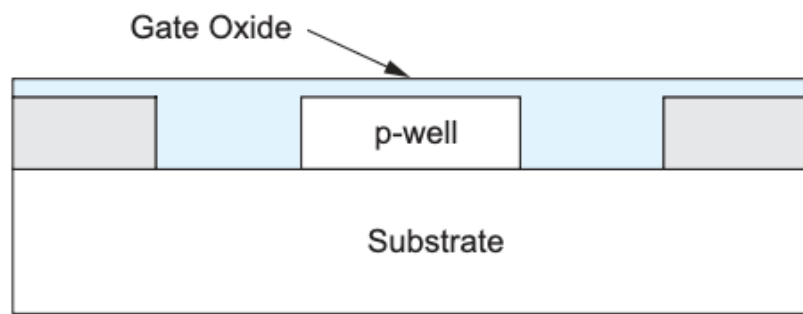


(d) CMP for Planarization

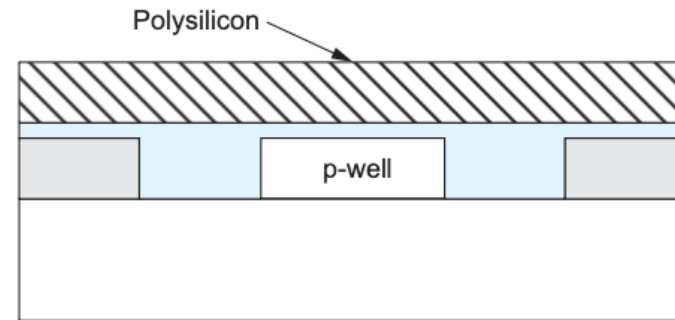
Gate Oxide Layer



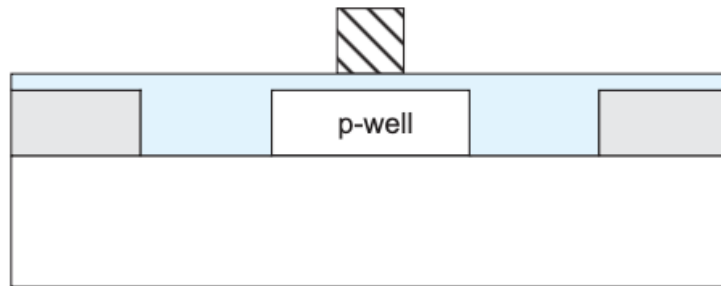
Gate and Source/Drain Formation



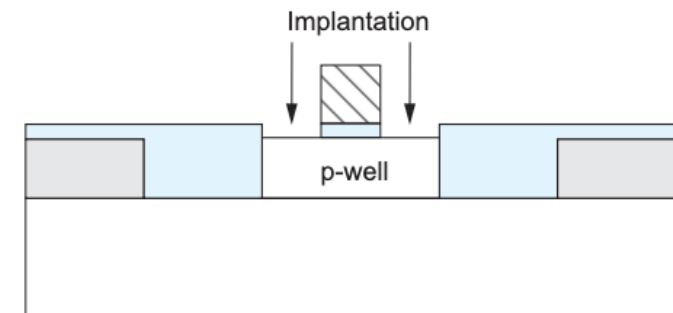
(a)



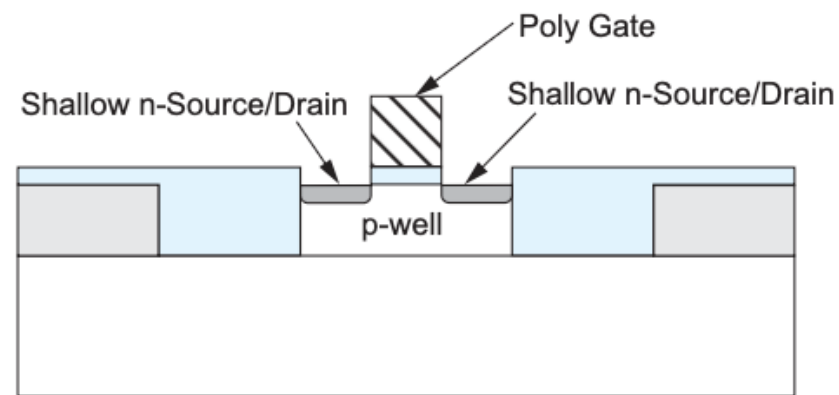
(b)



(c)

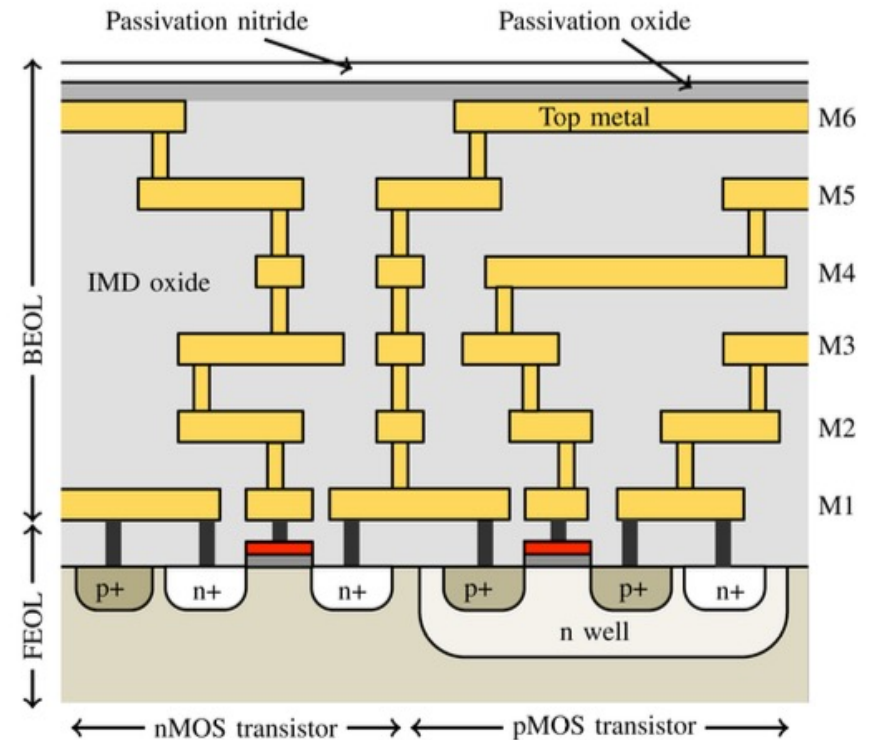
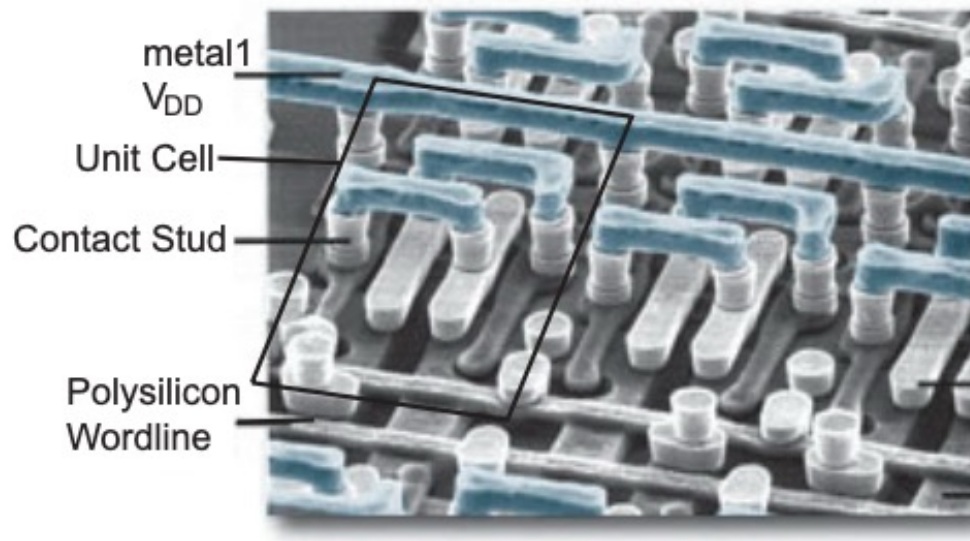
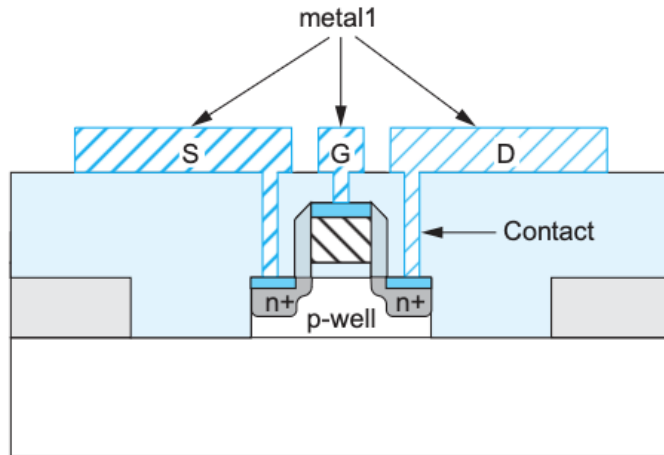


(d)



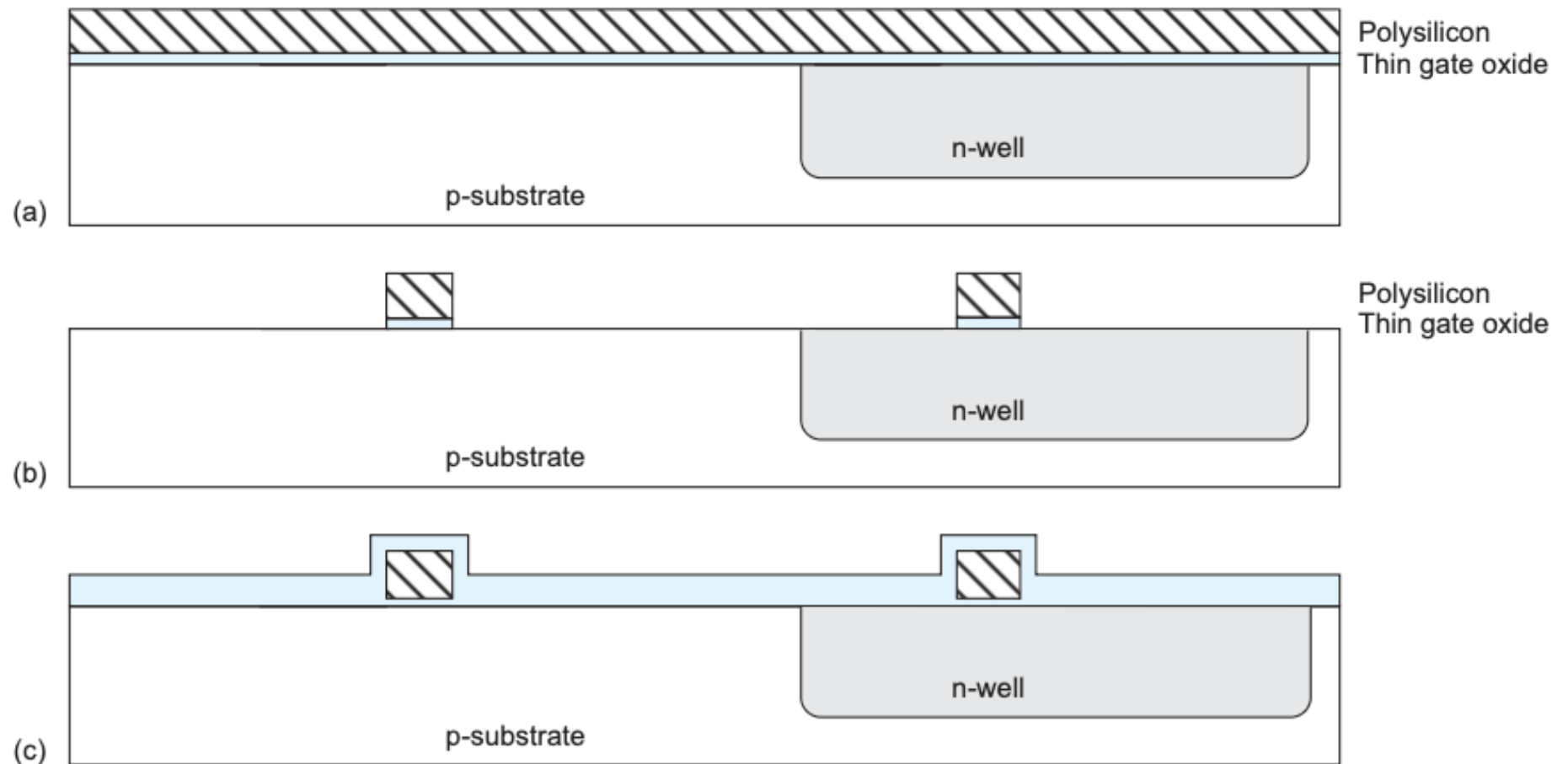
(e)

Contacts and Metalization



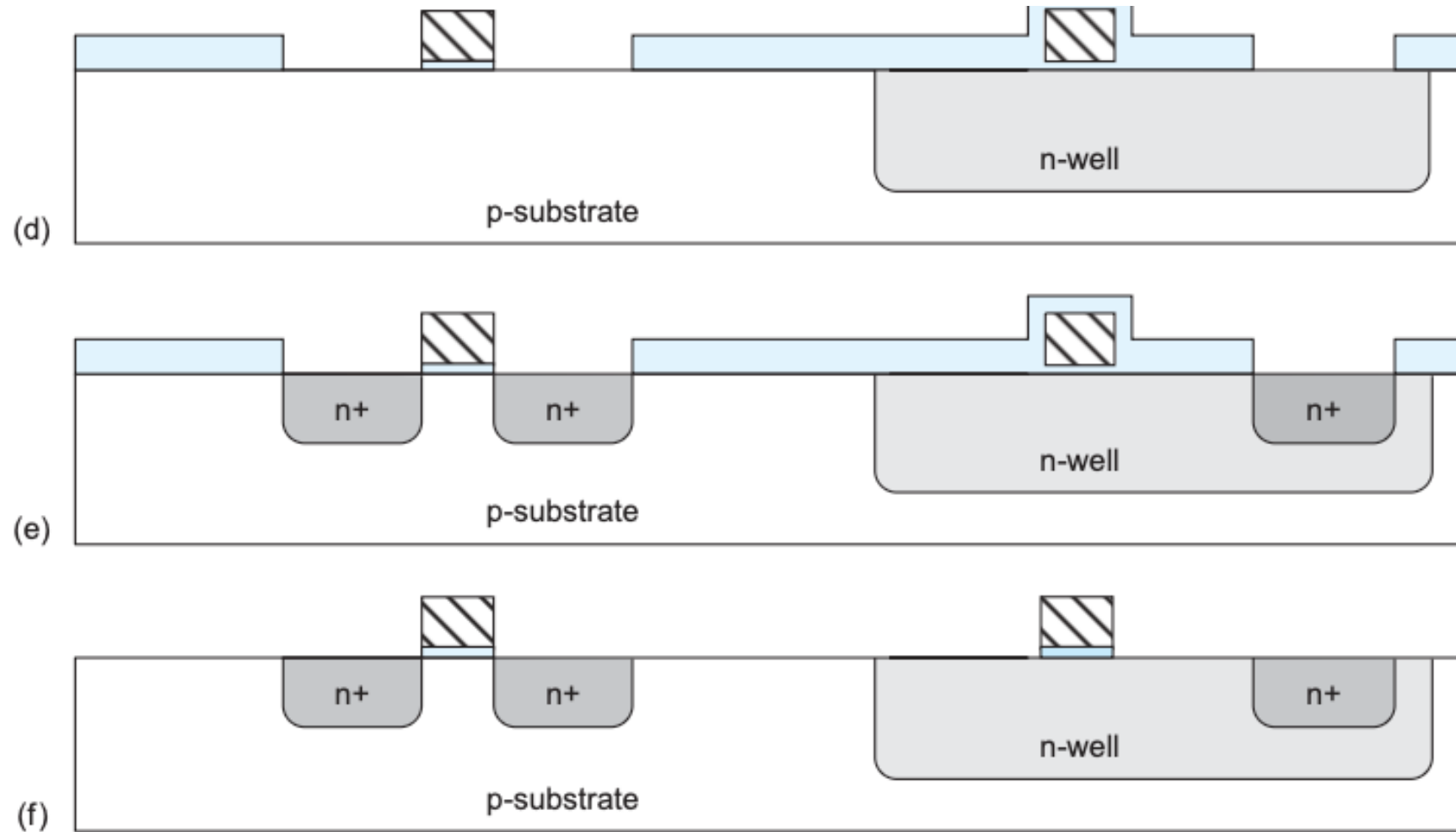
W&H p110-112

Putting all steps together (1)

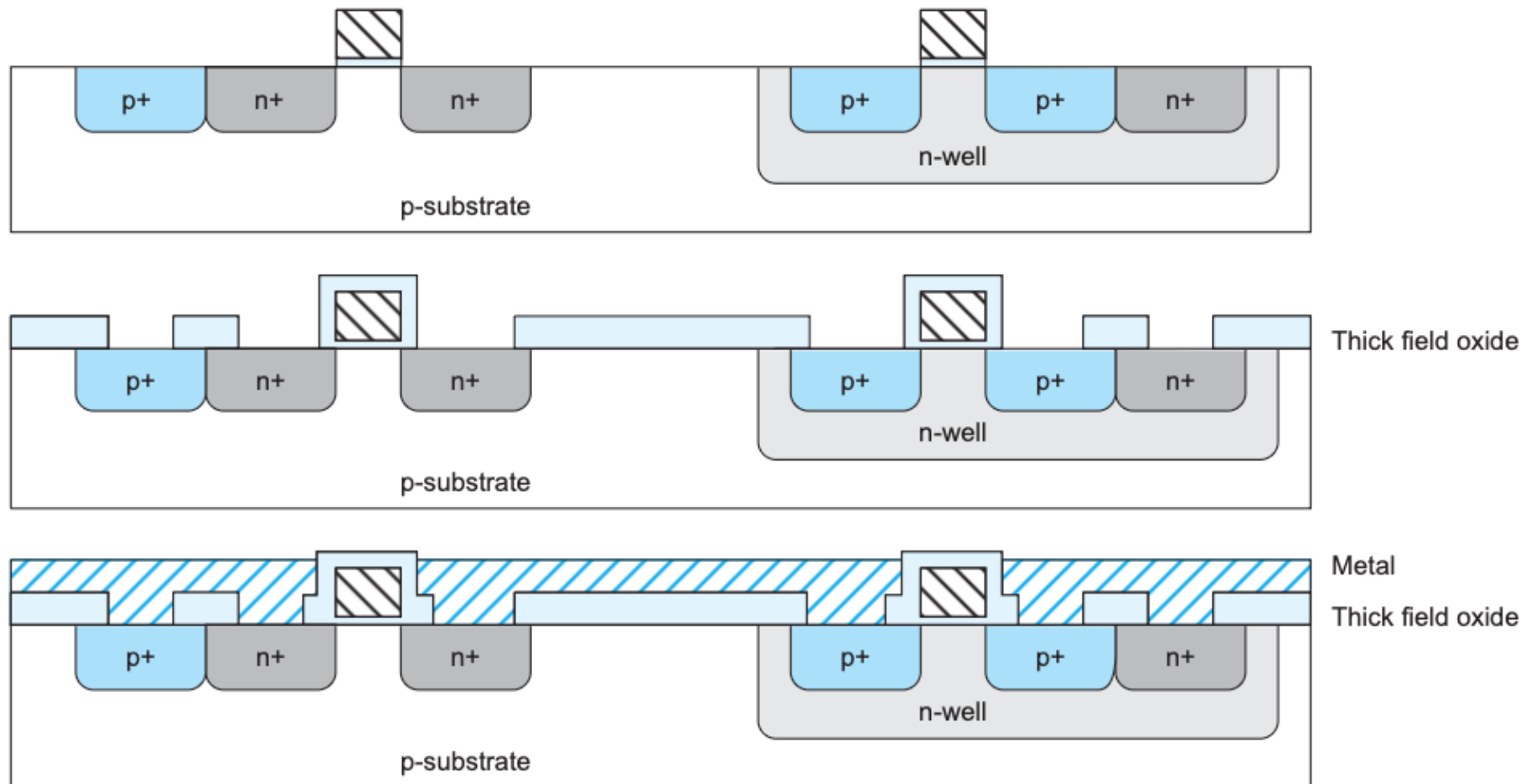


W&H p23-24

Putting all steps together (2)



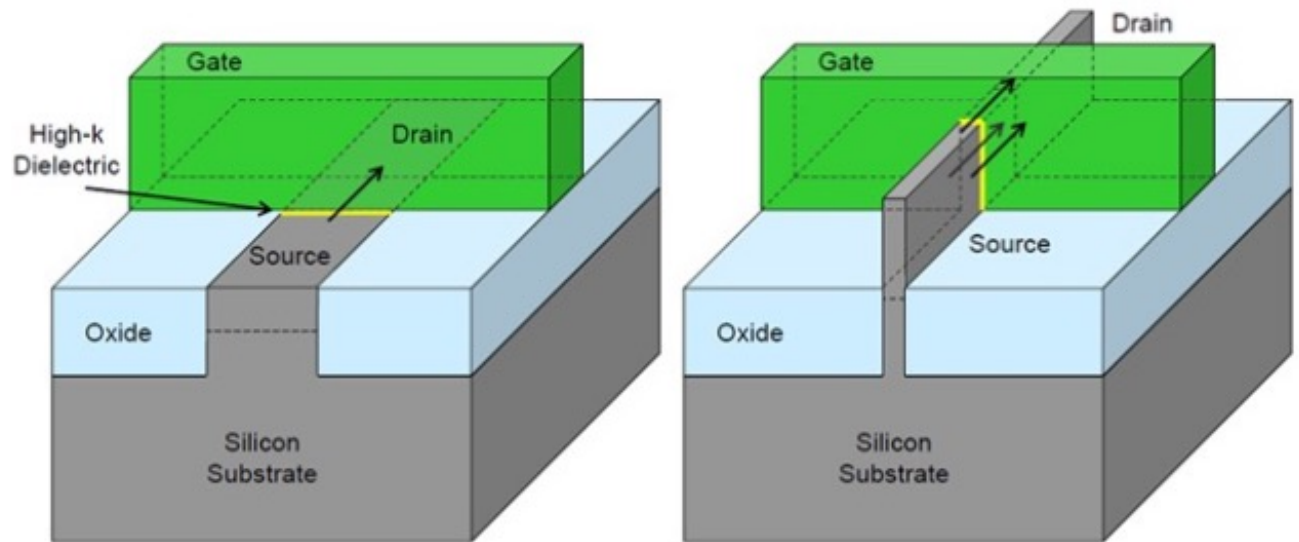
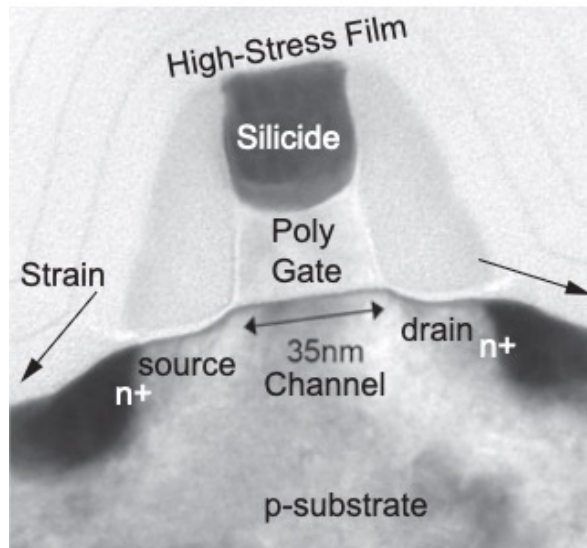
Putting all steps together (3)



W&H p23-24

Process Enhancements

- ◆ **Trench isolation (L3S17):** Shallow trench isolation (STI) prevents electrical current leakage between adjacent semiconductor device components.
- ◆ **High-K dielectrics/metal gate:** Replacing the SiO₂ gate dielectric with a high-K material allows increased gate capacitance without the associated leakage effects.
- ◆ **Strained silicon:** A layer of silicon in which the silicon atoms are stretched beyond their normal interatomic distance leading to better mobility, resulting in better chip performance and lowering energy consumption.
- ◆ **Gate engineering:** for within-die choice of multiple transistor threshold voltages (V_t) to optimize delay or power.
 - 3D MOSFET structure (FINFET) to improve controllability and reduce leakage.



Further Exploration

- ◆ Read the relevant part of Weste and Harris' book, part of Chapter 2 and 3.
- ◆ “How are chips made?” <https://www.youtube.com/watch?v=IkRXpFIRUI4>
- ◆ “Photolithography” <https://www.youtube.com/watch?v=TjpbrLTem3M>
- ◆ “TSMC Fab Tour” <https://www.youtube.com/watch?v=divOKxuYkIM>